QuickSort

Recap: QuickSelect

It is a **selection algorithm**. It looks for the **k-th smallest element** in an unsorted array.

It uses a **partition strategy** to find if index of the partitioned element is more than k, then we recur for the left part. If index is the same as k, we have found the k-th smallest element and we return. If index is less than k, then we recur for the right part.

Using a **random pivot** when defining partitions we can **avoid the worst case** (largest/smallest element picked as pivot).

Quick Select - Time Complexity

Best Case: when we partition the list into two halves and continue with only the half we are interested in.

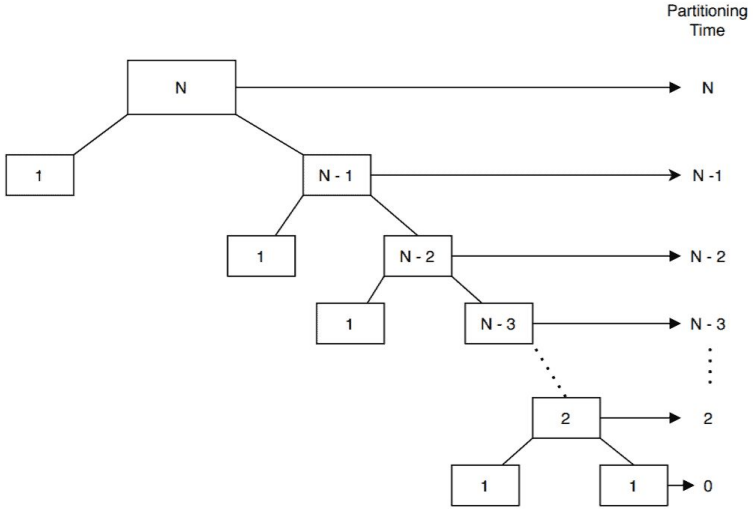
N + N/2 + N/4 + N/8 + N/16

This is a geometric series evaluates approximately 2 * N.

O(n)

Quick Select - Time Complexity

Worst case: larger/smallest element picked as pivot



O(N^2)

The **height of the tree** will be **N** and in **top node** we will be doing **N operations then N - 1** and so on until 1

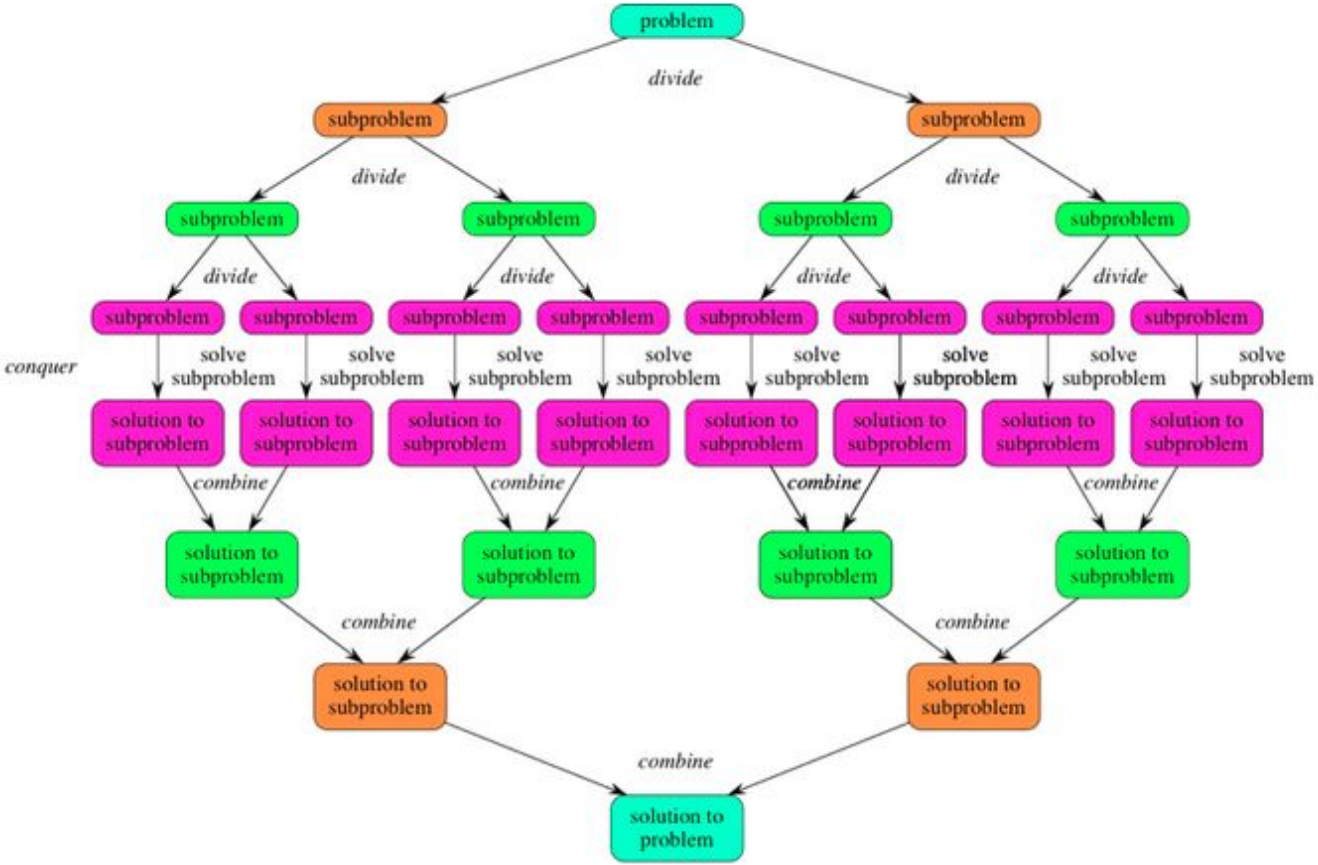
Divide and Conquer

Some algorithms, such as QuickSelect and QuickSort employ a common algorithmic paradigm based on recursion.

A divide-and-conquer algorithm recursively breaks down a problem into two or more sub-problems of the same or related type, until these become simple enough to be solved directly. (Wikipedia)

Divide and conquer algorithm parts

1. **Divide** the problem into a number of subproblems that are smaller instances of the same problem.
2. **Conquer** the subproblems by solving them recursively.
3. **Combine** the solutions to the subproblems into the solution for the original problem.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QuickSort

QuickSort uses the divide and conquer strategy (recursive algorithm).

- **Divide** by choosing any element (the pivot) and creating the partitions.
- **Conquer** by recursively sorting the subarrays array[start … pivot - 1] (all elements to the left of the pivot, which must be less than or equal to the pivot) and array[pivot + 1 ... end] (all elements to the right of the pivot, which must be greater than the pivot).
- **Combine** by doing nothing. The conquer step recursively sorts the entire array.

QuickSelect and QuickSort

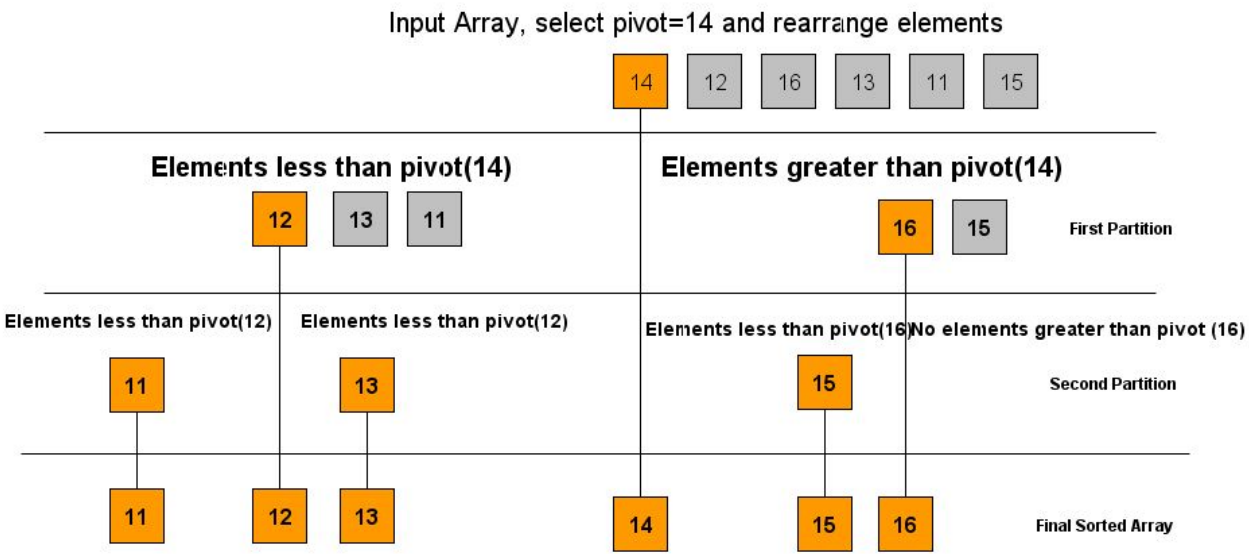
Both algorithms:

- It picks an element as a pivot
- It creates partitions based on the picked pivot

Difference:

- QuickSelect returns the value of the k-th element while the goal of the QuickSort is to use the logic partitions to sort the array

QuickSort



What does it happen after each call to partition?

QuickSort versions

A pivot element for the QuickSort can be picked in different ways:

- Pick median as the pivot
- **Pick a random element as a pivot:** preferred because it avoids a predictable pattern that could lead to the worst-case scenario.
- Always pick the first element as a pivot
- Always pick the last element as a pivot

Coding Time = Fun Time

Add the following method to your Quick.java file:

QuickSort

```
public static void quickSort(int[] data) {
        quickSort(data, 0, data.length - 1);
}

public static void quickSort(int[] data, int start, int end)
{
    // Your code here
}
```